Salient Object Detection

End-to-end CNN pipeline from scratch

- Input RGB image
- One-channel saliency mask
- Overlay for visual inspection

Dataset and Pipeline

ECSSD image/mask pairs with deterministic train/val/test splits

- Paired loader matches images and masks by filename stem
- Resize to 128 x 128 or 224 x 224
- Horizontal flip, random crop, brightness, and contrast jitter
- Masks stay aligned with nearest-neighbor resizing

CNN Architecture

From-scratch encoder-decoder with skip connections

- Four Conv2D encoder levels with ReLU, MaxPool, and BatchNorm
- U-Net-style skip connections preserve object boundaries
- ConvTranspose decoder upsamples back to 128 x 128
- Sigmoid output produces a one-channel saliency map

Results

Improved model beats the baseline and supports high precision

- Balanced mode: IoU 0.5364, F1 0.6915, recall 0.7711
- High-precision mode: precision 0.8726 at threshold 0.94
- Baseline F1 improved from 0.6278 to 0.6915
- Logs, checkpoints, metrics, and visual grids are saved

Demo and Next Steps

Upload image -> mask, overlay, and inference time

- Streamlit app loads the improved checkpoint by default
- Notebook shows the same inference workflow
- Report PDF and five-slide deck are included
- Repository is ready to share with the mentor